UNWRITTEN REFERENTS: SEPARATING AVAILABILITY FROM SCORING IN SELECTIVE ABSTENTION

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ABSTRACT

A language model that answers when it should abstain has failed at a decision, and in agentic and retrieval pipelines that failure is the input to the next step. When it happens, two different things may have gone wrong. The abstention score may be bad, or what the score needs may not be present where it is read. No evaluation we are aware of separates them. The knowledge-awareness signals that read model internals at entity positions are all read where the entity determining correctness is written down, and the sampling-based detectors that do not read positions at all cannot attribute their error to one either. Composed questions break that assumption. In *the mother of the performer of Imagine*, correctness depends on John Lennon, who is never named. We separate the two failures by manipulation, not by a better probe: write the deciding referent into the prompt, matched for length and entity popularity, and re-measure. The question, answer and label are unchanged, so performance that returns is availability and performance that does not is scoring. We decompose selective risk by which internal positions a rule may read and prove a floor no rule restricted to the written positions can beat, with both slack terms measured rather than assumed. On Gemma 2 2B a knowledge-awareness direction read at the written entity separates whether the first hop is known (AUROC 0.808) but is near chance on whether the composed query comes out right (0.574); writing the referent in recovers $+0.118$, and a length-matched distractor recovers nothing (0.531), so the effect is availability and not position. Availability is partly recoverable without touching the prompt. The same frozen direction read one token past the written entity clears a permutation null on all five models tested, and raises one-pass selective AUROC from 0.562 to 0.766 with a parameter-free rule. That is the best of any one-pass selector we ran, though two-pass Lookback Lens reaches 0.792; we report it as evidence that the availability term is recoverable, not as a detector to adopt. Two limits bound what this shows. The estimate rests on one composable relation pair, and no model in our experiments refuses on this template, so we measure a readout and the abstention built on it and not the refusal mechanism the direction was found by steering. We release the data, the code, and an appendix mapping every claim to the experiment supporting it.

1 INTRODUCTION

A model that answers when it should abstain has made two mistakes: a wrong answer, and the decision to give it. The second is the more consequential one in agentic and retrieval pipelines, where each step's subject is the previous step's output, so an unwarranted answer at step one is the input to step two. One line of work makes that decision from the model's own internal state: Ferrando et al. (2025) find linear directions separating recallable entities from unknown ones that causally control refusal, and Lindsey et al. (2025) reach a compatible account from attribution graphs. Both are read at the final token of an entity written in the prompt.

The limitation. That read location is not incidental, and recent work has made it the object of study. Orgad et al. (2025) show the choice of token is a general oversight and map probe accuracy

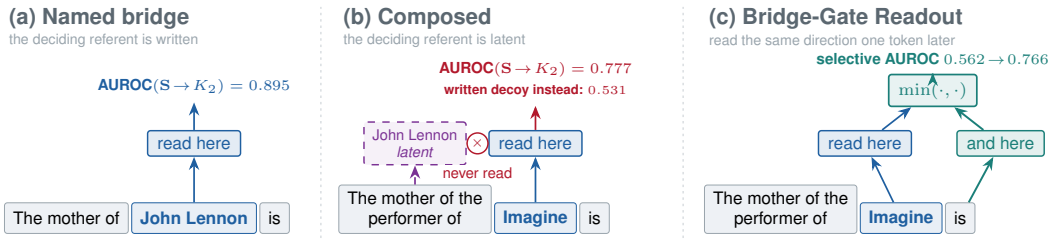


Figure 1: **The manipulation, and the recovery.** A knowledge-awareness direction (Ferrando et al., 2025; Lindsey et al., 2025) is read where an entity is written. **(a)** and **(b)** are the same question with the same answer and the same label, matched for length and entity popularity, differing only in whether the deciding referent is written. The performance **(a)** recovers over **(b)** is availability; what remains is scoring (Section 6.3). Writing a length-matched decoy instead of the true referent recovers nothing, so the effect is the referent and not the position. **(c)** The referent is resolved internally even in **(b)**, so reading the same frozen direction one token later and taking the minimum recovers part of the gap in one forward pass with no fitted combination rule (Section 4.3). Panels (a) and (b) report AUROC against K_2; panel (c) reports selective AUROC against Y. All numbers are Gemma 2 2B.

across layers and tokens; Luo et al. (2026) separate two truthfulness pathways by where their evidence originates; Wagner (2026) decomposes abstention into a correctness axis and an answerability axis and bounds what each signal recovers. All three ask which of the available positions to read; none can ask what happens when the entity deciding correctness occupies no position at all. So when a method fails, the literature cannot say whether the score was bad or whether what it needed was never written, and after Hussain & Kantarcioglu (2026) showed much of the reported progress here is benchmark artifact, that is the distinction the field now needs.

The insight. A manipulation settles this where a better probe cannot (Figure 1). Take a composed query, *the mother of the performer of Imagine*, whose answer depends on facts about John Lennon, an entity the prompt determines but does not name. Prior mechanistic work shows the model resolves this bridge referent internally, in early layers, before the second hop (Yang et al., 2024; Biran et al., 2024). Now write it in: *the mother of John Lennon*, matched for length and entity popularity. Same question, same answer, same label; only what is available to read has changed. Performance that returns is availability, performance that does not is scoring. We call a referent *latent* when the prompt determines it but does not tokenize it, the normal case in nested descriptions, multi-step lookups, and any pipeline where one step's output is the next step's subject.

The solution. We formalize the split as a decomposition of selective risk by *read set*, the (position, layer) pairs a rule may see, and the named-bridge comparison estimates the availability term.

Contributions.

1. **A decomposition of selective risk by read set, with a bound.** We split a rule's risk into an irreducible part, a part Δ_{blind} attributable to which internal positions the rule may read, and a part attributable to its scoring, then prove a floor on the first two that holds for any rule restricted to the written positions, with both slack terms measured rather than assumed (Theorem 1).
2. **A controlled measurement of the availability term.** The named-bridge comparison changes what is readable without changing the question, answer or label; a length-matched distractor controls for the position itself. Writing the referent in recovers $+0.118$ AUROC on downstream knowledge and the distractor recovers nothing, 0.531 against 0.895. To our knowledge this is the first estimate attributing a quantity of abstention error to information that was never written.
3. **Evidence that the availability term is recoverable, not merely diagnostic.** Reading the same frozen direction one token past the written entity raises one-pass selective AUROC from 0.562 to 0.766 ($p < 0.001$), the best of any one-pass selector we ran. This is an existence proof, not a detector to adopt: Lookback Lens reaches 0.792 at twice the cost

and our own BGR-lr reaches 0.781. What it shows is that the measured gap can be partly closed by reading elsewhere in the same forward pass.

The practical reading runs the other way from the diagnostic. A pipeline cannot write in a referent it does not yet know, but it can write down the ones it does resolve, and that puts them where an abstention signal can be read at all. Two limits belong here and not at the end: the availability estimate rests on one composable relation pair, and we do not establish *why* the gate fails to reach the latent position, because the models never refuse on this template (Section 6.6).

2 Related work

Knowledge awareness read at written entities. Ferrando et al. (2025) is the predecessor: sparse-autoencoder latents at the final token of a written entity separate recallable from unknown entities, causally steer refusal, and suppress attribute-extraction heads (Geva et al., 2023; Nanda et al., 2023). They also find uncertainty latents that predict incorrect answers without explicit refusal, so the signal is already known to reach beyond refusal behavior. Lindsey et al. (2025) reach a compatible account through attribution graphs over cross-layer transcoders (Ameisen et al., 2025; Marks et al., 2025), with refusal as the default and entity recognition as its inhibitor. Tamoyan et al. (2025) move the unit of analysis to the entity-relation pair. Every read-out in this line is taken at a position where the entity in question is written down.

Where to read, and what each read-out recovers. Three recent results make read location the object of study, not an implementation detail, and each constrains what we may claim. Orgad et al. (2025) show that the choice of token is a general oversight, map probe AUC across layers and tokens, find truthfulness concentrated at the exact answer tokens, and report that the resulting detectors do not transfer across datasets. Luo et al. (2026) separate two truthfulness pathways, one anchored in question-to-answer information flow and one deriving evidence from the generated answer, disentangle them by attention knockout and token patching, and tie both to knowledge boundaries. Wagner (2026) decomposes abstention into an answer-correctness axis and a question-answerability axis, shows a single confidence score conflates them, and measures each under surface controls with risk-controlled composition, including matched factual-relation gates on the same Gemma Scope dictionaries we use. Read location matters, it is measurable, and decomposing abstention into axes with per-axis controls is the right methodology; all three are already established. What none of them addresses is a position at which the deciding referent is fixed by the prompt and never tokenized, which is the only case we study. Our axis is orthogonal to Wagner's. His separates what the question is from what the answer would be, ours separates which positions a rule reads from how well it scores them.

Bounds, and why they are wanted now. Author list not confirmed (2026) treat hallucination onset as a change point, derive a delay floor, and split a detector's deficit into the information its features carry and the rule that accumulates it. That is the shape of Section 3's argument in a different formalism, on detection delay over a token stream rather than selective risk at a coverage, and neither implies the other. Hussain & Kantarcioglu (2026) supply the motive, reporting that progress here is substantially explained by benchmark construction artifacts. Selective prediction supplies the coverage-risk framework (El-Yaniv et al., 2010; Geifman & El-Yaniv, 2017; Chow, 1970), and the query-level uncertainty literature (Kadavath et al., 2022; Azaria & Mitchell, 2023; Burns et al., 2022; Chen et al., 2024; Ji et al., 2024; Gottesman & Geva, 2024; Kossen et al., 2024; Chen et al., 2026; Wen et al., 2025; Yao et al., 2025) scores a query as one object without asking where the score was read.

Latent multi-hop reasoning. Yang et al. (2024) found that models latently recall the bridge entity, and manipulated the prompt to mention it indirectly to raise internal recall; that targets recall of the answer, not the model's assessment of whether it can produce one. Biran et al. (2024) localized the two stages, Balesni et al. (2024) and Press et al. (2023) document the accuracy gap, and Yang et al. (2025b) separate composition from shortcut exploitation, which is what v measures and why our relation inventory follows their guessability screen. Liu et al. (2026) argue against hop-aligned circuits altogether, and their account predicts a readable bridge-position state with no gating pathway; we sweep positions rather than assume one and do not claim to exclude them. Lathkar (2026) bear on

our central manipulation: supplying one confirmed intermediate fact raises the confident-wrong rate at intermediate chain depths, so writing the bridge in is not a neutral intervention and we separate the level shift from the ranking change.

3 AVAILABILITY AND SCORING

The manipulation of Section 4.1 estimates a quantity. This section says which quantity, by separating "the score is bad" from "what the score needs is not there," and shows the second cannot be removed by improving the first.

Setup. A composed query is a chain $q = (e_1, r_1, \ldots, r_k)$ over a knowledge base (Vrandečić & Krötzsch, 2014), with head entity e_1, gold intermediate entities $b_1, \ldots, b_{k-1}$ where b_j is the object of (b_{j-1}, r_j) and $b_0 := e_1$, and target $a := b_k$. Alongside the composed prompt we form k single-hop prompts, the jth asking for the r_j of the *gold* b_{j-1}, not of whatever the model resolved, giving $Y = \mathbf{1}\{\text{composed prompt correct}\}$ and $K_j = \mathbf{1}\{\text{hop } j \text{ correct in isolation}\}$, scored by alias-aware exact match. Write $M := \prod_{j=2}^{k} K_j$ for the event that every hop after the first succeeds in isolation; at $k = 2$, M and K_2 are the same variable, so any discrepancy between numbers reported under the two names is an error.

A *read set* $\mathcal{R}$ is a set of (position, layer) pairs, and $g_{\mathcal{R}}$ is the *tuple* of per-position scores obtained by projecting the residual stream at each of them onto a frozen direction fixed in advance (Section 4.2). Reading it as a tuple rather than as one pooled scalar is what makes the nesting below work: if $\mathcal{R} \subseteq \mathcal{R}'$ then $g_{\mathcal{R}}$ is a sub-tuple of $g_{\mathcal{R}'}$, so $\sigma(g_{\mathcal{R}}) \subseteq \sigma(g_{\mathcal{R}'})$, which a single scalar summed over positions would not give. A selective rule answers when a score passes a threshold, at coverage c and selective risk $R = \mathbb{P}(Y = 0 \mid \text{answered})$ (Chow, 1970; El-Yaniv et al., 2010), and $R^*(Z, c)$ is the infimum of selective risk over rules seeing only Z. That infimum is attained by thresholding $\eta(Z) := \mathbb{P}(Y = 1 \mid Z)$ and accepting where it is largest, which is how we estimate it. Write $\mathbf{S}$ for the score at the written-entity position and $\mathbf{B}$ for the same direction at the bridge-resolution position, so that BGR reads the pair $(\mathbf{S}, \mathbf{B})$. We call $\mathbf{S}$ *the gate* throughout, following Ferrando et al. (2025), and reserve *referent* for the entity that determines correctness whether or not it is written.

Two sources of error. For nested read sets $\mathcal{R} \subseteq \mathcal{R}'$ and any rule s seeing only $g_{\mathcal{R}}$,

$$R(s, c) \;=\; \underbrace{R^*(g_{\mathcal{R}'}, c)}_{\text{irreducible}} \;+\; \underbrace{R^*(g_{\mathcal{R}}, c) - R^*(g_{\mathcal{R}'}, c)}_{\Delta_{\text{blind}}\text{: where it reads}} \;+\; \underbrace{R(s, c) - R^*(g_{\mathcal{R}}, c)}_{\Delta_{\text{subopt}}\text{: how well it scores}},$$

an identity, with both bracketed terms non-negative because an infimum over a larger feasible set is no larger and s is feasible in the second infimum. The algebra is not the contribution; the point is that abstention evaluations report only the sum, so reading in the wrong place and scoring badly are indistinguishable in the numbers the field publishes. Δ_{blind} is what the named-bridge comparison estimates and what BGR recovers.

A floor no rule at the written positions can beat. Δ_{blind} is relative to a reference read set; one absolute statement is available. In words: *if the written positions tell you nothing about whether the unwritten hops will succeed, and the model does not answer correctly by shortcut, no threshold on any score read there can push selective error below the rate at which those hops fail.* The two caveats are the only quantities needing definition.

Definition 1 (Slack terms)**.** *Essential suprema are taken with respect to the law of* $\mathbf{S}$*. The* leakage $\delta := \operatorname{ess\,sup}_s \big(\mathbb{P}(M = 1 \mid \mathbf{S} = s) - \mathbb{P}(M = 1)\big)^+$ *is how much the written-position score can overstate the chance that downstream hops succeed. The* shortcut rate $v := \operatorname{ess\,sup}_s \mathbb{P}(Y = 1, M = 0 \mid \mathbf{S} = s)$ *is how often a composed query comes out right anyway.*

Only the positive part is needed, and taking it matters. A gate that is *anti*-predictive of M, as two of our five are (Table 2), has a large two-sided deviation and a small one-sided one, so the two-sided definition would inflate the slack for no reason.

Theorem 1 (Written-position floor)**.** *Let a rule, randomized or not, accept on an event determined by* $\mathbf{S}$ *and an independent randomization, with positive probability of accepting. Then*

$$R \;\geq\; \mathbb{P}(M = 0) - \delta - v.$$

The proof (Appendix A) is four lines: such a rule cannot be more precise than the largest value of $\mathbb{P}(Y = 1 \mid \mathbf{S})$, which is at most $\mathbb{P}(M = 1) + \delta + v$ once that is split on M. Bounding the two pieces separately is slightly lossy and the code also reports the sharper single-supremum form; we keep two terms because leakage and shortcutting are tested by different controls.

Three consequences, each tied to an experiment. The bound holds unconditionally, since δ and v are definitions, not hypotheses, but is informative only when $\delta + v < \mathbb{P}(M{=}0)$, and Section 6.2 reports a regime where it is not. Both must be estimated conditionally on $\mathbf{S}$, since a marginal rate is not conservative but wrong, a trap the released smoke test caught on synthetic data (Appendix C). And comparing a chain with its own truncation gives $M_{k+1} \leq M_k$ pointwise, so $\mathbb{P}(M{=}0)$ is non-decreasing in k; whether the *bound* rises depends on $\delta + v$, which the $k{=}3$ condition answers and the theorem does not. The bound covers scores at the written positions, not arbitrary functions of the full state there, and is vacuous at $k = 1$. It is also vacuous on four of five models at $k{=}2$ (Section 6.2), which raises a fair question about what it earns. Two things. It identifies M as the quantity a written-position rule is up against, which is why the paper measures $\mathbb{P}(M{=}0)$, δ and v at all rather than only reporting AUROC. And it makes the failure mode checkable: a large measured δ would say the written positions do carry downstream information, and would refute the reading we give our own results.

4 METHOD

Three components in application order: an instrument that varies what is available to read, a frozen direction that reads it, and a rule combining two readings.

4.1 THE NAMED-BRIDGE INSTRUMENT

The instrument has one job, which is to change what is available to read while holding fixed everything the label depends on. Each composed item therefore yields a matched pair. The *composed* prompt is *The* $\langle r_2 \rangle$ *of the* $\langle r_1 \rangle$ *of the* $\langle$*type*$\rangle$ $\langle e_1 \rangle$ *is*; the *named-bridge* prompt is *The* $\langle r_2 \rangle$ *of the* $\langle$*type*$_2\rangle$ $\langle b_1 \rangle$ *is*, substituting the gold bridge for the description that determines it. Both ask for the same attribute of the same referent, carry the same label K_2, and end on the same token. The pair is matched on two dimensions, so the difference is availability and nothing else. **Length** is matched by padding the named-bridge prompt with a type-matched modifier to the composed prompt's token count, following the entity-late, fixed-ending form of Tamoyan et al. (2025). **Popularity** is matched by drawing the pair from the same 2024 Wikipedia pageview decile, so a gap cannot be produced by the bridge being a more prominent entity than the description implies.

The instrument needs one negative control, because naming *any* entity at that position might help by giving the direction a concrete token to fire on. The *distractor* prompt substitutes an unrelated real city of matched token length for b_1 in the named-bridge template. If the gap is availability, the distractor recovers nothing.

4.2 GATE EXTRACTION: FIT ONCE ON ONE-HOP DATA, THEN FROZEN

The direction w is fit on *one-hop* prompts only, on the fit split, and never refit. A direction refit on composed data would show only that some linear readout exists; a frozen direction that transfers to a position where nothing is written shows the state is expressed in the same format. Splits are by *entity*, not by prompt, so no entity appears on both sides under a different relation.

Three estimators are computed so that no single one is load-bearing (Kissane et al., 2024): a separation-score sparse-autoencoder latent (Lieberum et al., 2024; He et al., 2024), a difference-of-means direction, and an ℓ_2-regularized logistic probe (Appendix F). Pairwise Spearman agreement ≥ 0.50 on held-out single-hop prompts, together with AUROC ≥ 0.65 for each, was pre-registered as the precondition for reporting anything downstream; Section 6.1 reports that no model meets it across all three, and that we proceed on the two that agree.

4.3 BRIDGE-GATE READOUT

If a latent referent is resolved internally, the same frozen w read at the position where that happens should carry part of what writing it in would have supplied. Let $g_{\mathbf{S}}$ be the score at the final token of the written entity and $g_{\mathbf{B}}$ the score at the bridge-resolution position. At inference, on one forward pass:

1. Read $g_{\mathbf{S}}$ and $g_{\mathbf{B}}$ from the residual stream at the fit layer.
2. Standardize each to zero mean and unit variance using statistics computed on the fit split, not on the evaluation items.
3. Score $s = \min(g_{\mathbf{S}}, g_{\mathbf{B}})$ and abstain when $s < \tau$.

Why the minimum. Answering a composed query correctly requires *both* hops, so the natural aggregator is a conjunction, and on standardized scores the minimum is its soft form: one low reading is enough to abstain, and a high surface reading cannot compensate for a low bridge reading. Sum or mean would allow exactly that compensation. This is a design choice with a testable consequence, so we report min against mean and max in Appendix F.

Position selection, and the fairness problem it creates. The bridge-resolution position is the offset maximizing frozen-direction AUROC in the sweep of Section 6.4. Our method therefore gets a searched read position while every internal-state baseline is run at whichever position its own paper specifies. That asymmetry, not the aggregation rule, is the most likely explanation a skeptical reader will reach for first, and the comparison in Table 9 does not currently rule it out.

BGR-lr replaces step 3 with a two-feature logistic regression on $(g_{\mathbf{S}}, g_{\mathbf{B}})$, cross-fitted so that no item contributes to the model that scores it. BGR-min is the headline method because its *combination rule* has no fitted parameters and therefore cannot absorb the effect into a learned model. It is not unsupervised: the direction w is fit on one-hop known-versus-unknown labels and the standardization uses fit-split statistics, so BGR belongs with the supervised probes in any fair grouping, not with semantic entropy or SelfCheckGPT. Table 9 groups by inference cost, which hides this.

Cost. Both variants read two activations from a forward pass the model already performs, so the marginal cost is two dot products, against two full passes for Lookback Lens and ten samples for INSIDE.

5 EXPERIMENTAL SETUP

The instrument, the direction and the rule are all model-agnostic; what follows is what we ran them on and what we measured.

Models and data. The sparse-autoencoder analysis is confined to models with public dictionaries: Gemma 2 2B and 9B, base and instruction-tuned (Team et al., 2024), with Gemma Scope (Lieberum et al., 2024), and Llama 3.1 8B (Grattafiori et al., 2024) with Llama Scope (He et al., 2024). Difference-of-means and probe estimators need no dictionary, so those extend to Llama 3.1 70B and Qwen3 32B (Yang et al., 2025a) in 4-bit.

Data. Composed and matched one-hop prompts come from the Wikidata-derived sets of Yang et al. (2024) and Biran et al. (2024), a controlled set over the entity types of Ferrando et al. (2025) so gate directions transfer without domain shift, and natural-language checks on MuSiQue (Trivedi et al., 2022) and 2WikiMultiHopQA (Ho et al., 2020). Correctness is alias-aware exact match against Wikidata. Three families of latent referent: F1 bridge entities, F2 three-hop chains for the compounding clause, and F3 self-written entities, where the model writes the bridge into its chain of thought. Every item ships with its one-hop decompositions, so K_1, K_2 and Y need no human annotation (Table 7, Appendix F).

Further controls. Beyond the distractor of Section 4.1: Wikipedia-derived multi-hop sets overstate composition because models memorize composites, exactly what v measures, so we report v and evaluate counterfactual chains; and alias-aware exact match can under-count unusual surface forms, so we validate it against an LLM judge (Zheng et al., 2023). Protocols in Appendix F.

Metrics. AUROC against Y and against K_2, AURC, selective risk at 50% coverage, $\hat{\delta}$, $\hat{v}$, $\hat{\Delta}_{\text{blind}}$, and forward passes per scored item. Intervals are percentile bootstrap over queries; AUC comparisons use a paired bootstrap and DeLong (DeLong et al., 1988).

6 RESULTS

Numbers come from a real run on Gemma 2 2B (base), layer 8. One weakening shapes everything after it: every model here produces a refusal rate of exactly 0.000 on this template at every steering coefficient tried (Appendix B, control (f)), so every claim below concerns a linear readout of the gate direction and the abstention built on it, not the model's own refusal. Appendix E maps each claim to its experiment and marks those still open, the most consequential being a bridge-only arm that would show whether BGR's gain comes from the conjunction or merely from the extra read position.

6.1 POSITIVE CONTROL, THE GATE WORKS WHERE IT SHOULD

Nothing downstream is interpretable if the direction does not separate known from unknown entities where they *are* written, so that check runs first. On Gemma 2 2B at layer 8, difference-of-means (AUROC$=0.886$, CI $[0.876, 0.896]$) and the probe (AUROC$=0.887$, CI $[0.858, 0.885]$) separate known from unknown entities on held-out single-hop prompts, both above the pre-registered 0.65 threshold. Prediction of K_2 from the gate at the *named* bridge is AUROC$=0.895$/0.929 (diffmeans/probe), so downstream knowledge is gate-visible when the entity is written, the premise whose absence the rest of the paper tests.

The agreement half needs stating exactly. The pre-registered criterion was agreement across all three estimators and no model meets it: the 0.547 above excludes the sparse-autoencoder estimator, which sets the minimum pairwise agreement at 23 of 26 layers in the refit sweep (Appendix B, control (d)). Four models clear a two-estimator gate on that relaxed reading and Qwen3-32B clears neither (Table 3). Relaxing a pre-registered criterion after seeing the data is a move a reader may discount, so we record it instead of four clean passes. The defence is specific: the two estimators we use reach AUROC 0.75–0.91 on every model and agree with each other, and the one that fails produces none of the numbers below. "The gate" below means those two.

6.2 THE GATE SEPARATES THE WRITTEN HOP, NOT THE COMPOSED OUTCOME

On $n = 239$ held-out composed queries with $K_1 = 1$, the gate reaches AUROC$(\mathbf{S} \to K_1) = 0.808$, AUROC$(\mathbf{S} \to M) = 0.616$ and AUROC$(\mathbf{S} \to Y) = 0.574$. It separates whether the written entity's own hop is known and is close to chance on whether the composed query will come out right. That contrast, within one set of queries and one score, is the finding: the shortfall is not that the gate is uninformative, but that what it is informative about is not what decides the answer. Table 2 (Appendix B) repeats the check on four more models spanning a $35\times$ range. Only Gemma 2 2B has AUROC$(\mathbf{S} \to Y)$ measured, so what the table can support is the weaker ordering AUROC$(\mathbf{S} \to K_1)$ >AUROC$(\mathbf{S} \to M)$, which holds on all five with gaps of 0.19 to 0.45. Those gaps show no trend with parameter count (Spearman $\rho = 0.10$ over five models, which five points cannot resolve either way), so we say the effect does not shrink with scale in our range rather than that scale is irrelevant to it.

Four things that cut against this reading. We group them instead of attaching one to each number. **(i)** The floor is vacuous here: at coverage 0.5, $\widehat{\delta} = 0.179$, $\hat{v} = 0.400$ and $\mathbb{P}(M{=}0) = 0.439$, so Theorem 1 evaluates to -0.140. A negative lower bound on a risk says nothing, and the shortcut term is why. At the gate value where it is worst, two in five composed queries come out right even though a downstream hop fails in isolation; $\hat{v}$ is an essential supremum over gate values, not a rate over all queries. The bound becomes informative only at $k{=}3$. **(ii)** $\widehat{\Delta}_{\text{blind}}$ is exactly 0.000 on Llama 3.1 8B and Qwen3-32B, and Llama 3.1 70B's 0.049 sits inside its own sampling-noise tolerance (Section 6), so it is separately estimable on two of five models. **(iii)** AUROC$(\mathbf{S} \to M)$ was pre-registered at ≈ 0.50; Gemma 2 9B gives 0.375 and Llama 3.1 8B 0.320, informative in reverse. A sign-flipped readout might recover part of that; we did not test it. **(iv)** We pre-registered an overlap check against single-hop queries the model answers correctly, expecting overlap; the distributions

are far apart ($D = 0.637$, $d = -1.415$), so it fails. The gate registers that a query is composed. What it cannot do is rank composed queries against each other, the 0.574 above.

And two about the sample. The $K_1{=}1$ filter runs without the Patchscopes (Ghandeharioun et al., 2024) confirmation the design called for, since no layer among three tried clears both the agreement bar and a working decode. And the class balance of the 239 items is not reported in our own outputs, so Appendix D infers it from control (n) before any interval can be stated; a paper that cannot state the positive rate of its own evaluation set has a gap, and we name it.

6.3 Writing the referent in: how much of the gap is availability?

The contrast above is consistent with two accounts that matter differently. Either the gate scores badly on composed queries, or what it needs is not at the position it reads. The named-bridge comparison separates them. Each composed prompt is paired with a prompt that names the same bridge entity directly, matched for length by a type-matched modifier and drawn from the same pageview decile, so the question, the answer and the label are unchanged and only what is written differs.

On the two models where we ran the comparison, writing the referent in recovers a substantial part of the gap. On Gemma 2 2B, AUROC($\mathbf{S} \to K_2$) rises from 0.777 composed to 0.895 named, a gap of $+0.118$; on Llama 3.1 70B it rises from 0.637 to 0.793, a gap of $+0.155$. The two gaps are not distinguishable at these sample sizes, since the difference is 0.037 against a 95% interval of roughly ± 0.094 (Appendix D), so we report both and claim no scale trend. The gate is not scoring badly. It is reading a position from which the deciding referent is partly invisible.

The obvious alternative is that naming *anything* there helps, by giving the gate a concrete entity token to fire on. It does not: substituting a length-matched unrelated city for the true bridge collapses AUROC($\mathbf{S} \to K_2$) to 0.531, chance, against 0.895. The gate reads the properties of the referent written, not the presence of one. Attention and counterfactual checks guard the reading. Attention from the final prompt token to the bridge position is 0.112, $1.82\times$ what an ignored position would receive, so the model is not simply looking elsewhere; and substituting a false premise moves the model to the stated premise rather than the memorized fact ($p_{\text{composed}} = 0.732$ against 0.021), so memorized composites are not carrying the result. Protocols in Appendix B.

One caveat bears on how the gap should be read. Lathkar (2026) report that supplying a confirmed intermediate fact can raise a model's confidence on answers it gets wrong, so writing the bridge in shifts the level of the score as well as its ranking, and only the ranking change is what availability means here.

6.4 Reading the referent where it resolves

Writing the referent in is a diagnostic, not a deployment option, because in a real pipeline nobody knows the bridge in advance. Can any of that availability be reached without changing the prompt?

Frozen-direction AUROC at three offsets around the bridge, at the layer where each model's gate is trustworthy, has all five peaks clearing their permutation nulls comfortably (mean-max ≈ 0.51, $p < 0.001$). Gemma 2 2B peaks at 0.788 at offset $+1$, where four of five models peak; Qwen3-32B peaks at offset 0 (Figure 3). The state the gate would need is linearly present one token past the written entity. That is weaker than Patchscopes confirming a resolved entity is *written* there, and weaker again than any claim about how the model uses it, but it is enough to motivate reading there.

6.5 Recovering availability without rewriting the prompt

Table 1 compares BGR against the strongest baseline in each inference-cost tier; Table 9 gives all thirteen.

Table 1: **Selective abstention on composed queries, best per inference-cost tier.** Gemma 2 2B, $n{=}239$. Grouped by inference cost. The comparison BGR wins is within the one-pass tier and not across the table, since the highest AUROC anywhere here belongs to a two-pass method. Best per column within a tier in bold. All values are single-run point estimates; only the BGR-versus-surface-gate difference has been tested (paired bootstrap and DeLong, $p < 0.001$).

Selector	Cost	Selective prediction		
		AUROC ↑	AURC ↓	Risk @ 50% ↓
One forward pass				
Surface gate (Ferrando et al., 2025)	1×	0.562	0.184	0.160
BGR-min (parameter-free rule)	1×	0.766	0.093	**0.084**
BGR-lr	1×	**0.781**	**0.079**	**0.076**
Extra passes or sampling				
Lookback Lens (Chuang et al., 2024)	2×	**0.792**	**0.089**	**0.084**
INSIDE (Chen et al., 2024)	k=10	0.596	0.116	0.134
Multi-call decomposition				
Self-ask (Press et al., 2023)	multi-call	0.500	0.188	0.227
Reference, not a ceiling				
Oracle observing M	–	0.725	0.095	0.076

The oracle observes M directly and is a reference point, not an upper bound: a binary oracle scores worse on AUROC than a continuous selector can, which is why BGR appears above it. Lookback Lens ties BGR-min on risk at 50% coverage and beats both variants on AUROC at twice the cost. Full results for all thirteen baselines are in Table 9, Appendix F.

BGR-min and BGR-lr beat the surface-gate baseline by $+0.203$ and $+0.218$ AUROC (paired bootstrap and DeLong both $p < 0.001$), the paper's one clean quantitative result and the only comparison here that survives even an unpaired interval at this sample size (Appendix D). Of the thirteen baselines in Table 9, Lookback Lens (Chuang et al., 2024) has the highest AUROC of anything we ran including our own methods (0.792 against 0.781 and 0.766), at twice the cost and with a worse AURC than BGR-lr. The claim BGR supports is therefore about the one-pass tier, not about the field. $P(\text{True})$ (Kadavath et al., 2022) at 0.623 and INSIDE (Chen et al., 2024) at 0.596 are next; everything else underperforms, down to 0.414. Self-ask and Operadic consistency (Bottman et al., 2026) sit at exactly chance because the 0.000 refusal rate collapses both to a constant score, a shared consequence of that null rather than a failure of either method. On Llama 3.1 70B ($n = 494$) BGR still beats the surface gate (0.666/0.669 against 0.584, $p \leq 0.001$), at less than half the margin.

What weakens the comparison. The reference row is *our* reimplementation of the gate of Ferrando et al. (2025), not a number they report, and at 0.562 it is close enough to chance that underimplementation is a live alternative reading.

6.6 What did not work

Availability says what a rule can read, not why the gate fails to reach the latent position. Four interventions were designed to answer that and none does, for the same reason each time: bridge patch moves recall by $\Delta_{\text{recall}} = -0.047$ against its own standard deviation of 0.212, and the refusal arms return 0.000 under intervention because the baseline refusal rate is 0.000 too, so there is no live signal to move. The double dissociation the design was built around has one weak arm and one inert arm, so we do not claim the causal test passed. What remains is correlational, and it does not exclude Liu et al. (2026), under whose account the same bridge-position readout would exist with no gating pathway to be blind. Protocol and per-intervention numbers in Appendix B. Chain of thought, which should on our account restore what the named-bridge condition restores, points that way but rests on 44 and 46 correct-arm items whose intervals include chance, so it is a direction and not a result.

Three limits follow. With no observable refusal we measure a readout and the abstention built on it, not the mechanism Ferrando et al. (2025) and Lindsey et al. (2025) steer. The availability estimate rests on one relation pair, place-of-birth → country, which Yang et al. (2025b) exclude from their own two-hop data because the bridge is inferable from the head entity's surface form; it is therefore the relation most likely to overstate composition, and $\hat{v} = 0.400$ is what that looks like from the inside. And three measurements would change our reading: a large $\hat{\delta}$ on a wider inventory, a bridge-only arm matching BGR-min, or a named-bridge gap that vanishes once relations are screened for guessability. The second is one line of analysis on data we already have.

7 CONCLUSION

A knowledge-awareness direction is read at tokens, while correctness depends on referents, and the two coincide only while every entity that matters is written down. When they come apart, an abstention score built on that direction may be scoring badly or may be reading a position from which the deciding referent is invisible. Writing the referent in separates the two, and on the two models where we ran that comparison a measurable share of the shortfall is availability rather than scoring. Part of it is recoverable without touching the prompt, by reading the same frozen direction where the referent resolves. We do not show that the model uses this signal to decide whether to answer: no model in our experiments refuses on this template, so the link to refusal behavior is untested here and is the first thing a follow-up should establish. What the results do support is narrower and still useful. An abstention result should state which positions its score was read from, because two methods reporting the same AUROC may be failing for different reasons; and a pipeline that keeps its intermediate referents out of its own context keeps them out of reach of any signal read at those positions.

REPRODUCIBILITY STATEMENT

All datasets are public and derived from Wikidata, and all models and sparse autoencoders are open-weight. Correctness labels are computed by alias-aware exact match, so no human annotation enters the pipeline and the labeling is exactly reproducible. Reproducibility is uneven and we would rather say where. Data construction, labels, gate fitting and metric computation are fully specified. Three implementation choices are not recoverable from the text as it stands: which estimator produces the reported gate score, how the two BGR scores are standardized, and which split the read position was selected on. Each is flagged where it occurs. Section 3 states every assumption (δ, v) as a measured quantity, not a supposition, and Appendix A gives complete proofs. Gate directions are fit on one-hop data and frozen, with the fitting protocol, layers, and hyperparameters in Appendix F. Prompt templates are in Appendix G. Code implementing every metric, with a synthetic smoke test verifying Theorem 1 numerically, is released with the paper. Bootstrap counts (2,000 resamples) and correction procedures are stated in the setup.

ETHICS STATEMENT

This work studies when models fail to recognize the limits of their own knowledge. Any signal that supports abstention can be inverted to suppress it, as Ferrando et al. (2025) note for the entity gate, and the direction we read is the same one they released, so the marginal risk from this paper is the read position rather than the direction. We judge the release net positive because the primary use is reliability and the underlying directions are already public.

One risk is specific to reporting a method as improving abstention. Our numbers come from one composable relation pair, and a threshold calibrated here and deployed as though it transferred would produce over-confident abstention in exactly the agentic settings the introduction motivates. Appendix E states which claims hold on how many models, and we would not want the method used before the generality experiments listed there have run. No human subjects, no personal data, and no model training beyond linear probes; entities are public figures drawn from Wikidata.

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A PROOFS

Proof of Theorem 1. Let $\kappa := \operatorname{ess\,sup}_s \mathbb{P}(Y = 1 \mid \mathbf{S} = s)$ and let A be the acceptance event, which lies in $\mathcal{G} := \sigma(\mathbf{S}) \vee \sigma(U)$ and has $\mathbb{P}(A) > 0$. Since $U \perp (H, Y)$, $\mathbb{P}(Y = 1 \mid \mathcal{G}) = \mathbb{P}(Y = 1 \mid \mathbf{S}) \leq \kappa$ almost surely, so randomization does not raise the ceiling and the argument covers randomized rules. By the tower property,

$$\mathbb{P}(Y = 1 \mid A) = \frac{\mathbb{E}[\mathbf{1}_A \, \mathbb{P}(Y = 1 \mid \mathcal{G})]}{\mathbb{P}(A)} \leq \kappa, \qquad \text{hence} \qquad R \geq 1 - \kappa.$$

It remains to bound κ. Partitioning on M and discarding nothing,

$$\begin{aligned} \mathbb{P}(Y = 1 \mid \mathbf{S}) &= \mathbb{P}(Y = 1, M = 1 \mid \mathbf{S}) + \mathbb{P}(Y = 1, M = 0 \mid \mathbf{S}) \\ &\leq \mathbb{P}(M = 1 \mid \mathbf{S}) + \mathbb{P}(Y = 1, M = 0 \mid \mathbf{S}), \end{aligned}$$

and by Definition 1 the first term is at most $\mathbb{P}(M = 1) + \delta$ almost surely, since δ bounds the positive part of $\mathbb{P}(M = 1 \mid \mathbf{S}) - \mathbb{P}(M = 1)$, and the second is at most v. Hence $\kappa \leq \mathbb{P}(M = 1) + \delta + v$ and $R \geq \mathbb{P}(M = 0) - \delta - v$. The two suprema need not be attained at the same value of $\mathbf{S}$, so $\varepsilon := \operatorname{ess\,sup}_s[(\mathbb{P}(M = 1 \mid \mathbf{S} = s) - \mathbb{P}(M = 1))^+ + \mathbb{P}(Y = 1, M = 0 \mid \mathbf{S} = s)] \leq \delta + v$ gives the sharper bound $R \geq \mathbb{P}(M = 0) - \varepsilon$ by the same argument. □

Scope. This is the only formal result in the paper. The decomposition of Section 3 is an arithmetic identity and is not proved here. The theorem's work is identifying M as the binding quantity and making δ and v measured rather than assumed, so that a claim about what a mechanism cannot do becomes falsifiable: a large measured δ would refute the reading this paper gives its own results. It constrains only scores measurable with respect to $\mathbf{S}$, not scores built from sampling, other internal states, or retrieval, which is why we test baselines outside that class rather than claim the result covers them. A machine harness checks the bound and the monotonicity remark numerically against randomly drawn joint distributions and finds no violation. It also confirms that the infimum defining R^* is attained by thresholding $\eta(Z) = \mathbb{P}(Y = 1 \mid Z)$, which is what licenses the plug-in estimator, and that the non-negativity of Δ_{blind} requires the tuple-valued read of Section 3: with one scalar pooled over positions, a larger read set does not refine the smaller one and the term can go negative by construction. None of that is a derivation, and the flag above still stands.

B CONTROL SUITE

Controls are listed with their status, not only their results, because several did not come out as designed. The suite has three jobs: rule out explanations of the availability gap that do not involve availability (a, c, e, g, i), rule out benchmark artifacts (h, n), and test whether the effect is specific to one model, layer, entity type or relation (b, d, j, k, l). The four interventions of Figure 2 were designed to establish a causal account and did not (Section 6.6); they are retained here so that the attempt and its failure are on the record rather than absent from it.

Four interventions separate the competing explanations, specified in full in Table 8, Appendix B. Surface steer steers the unknown-entity direction at the surface position and asks whether the induced refusal is selective on K_2. Bridge patch is the decisive behavioral test. It patches the bridge representation from a matched query with an unknown bridge, and measures the change in recall and in refusal on the same queries. A change in recall with no matching change in refusal is a double dissociation. Gate inject injects the unknown-entity direction at the *bridge* position, distinguishing a missing routing pathway from an unused one, which decides whether a repair must add routing or only supply a signal. Graph trace makes "not routed" a directly measured property of the computational graph rather than an inference from behavior, using attribution graphs over transcoders (Ameisen et al., 2025; Hanna et al., 2025). Following Lindsey et al. (2025), every graph-derived claim is validated by intervention in the underlying model, because the graph describes a replacement model.

Controls, protocol and status. (a) Named-bridge control, where the gap must close. **Run**, the gap closes and reverses direction on both models tested, Gemma 2 2B (AUROC($\mathbf{S} \to K_2$) $= 0.777$ composed vs. 0.895 named-bridge, gap $+0.118$) and Llama 3.1 70B (gap $+0.155$, composed 0.637

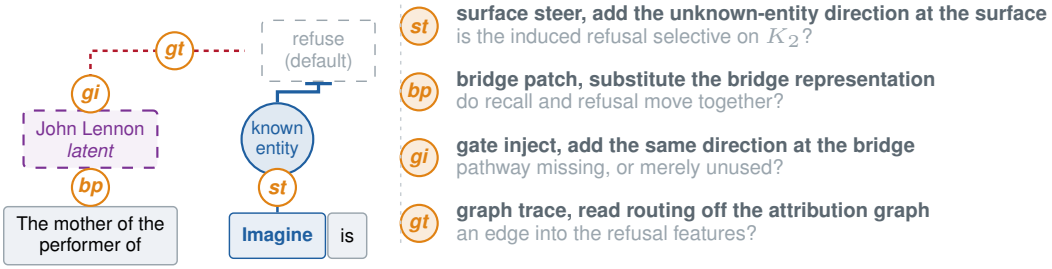


Figure 2: **The causal design.** All four interventions act on the same circuit, shown once at left with each marked at the position it targets. Surface steer and gate inject add the unknown-entity direction at the surface and bridge positions, bridge patch substitutes the bridge representation from a matched query with an unknown bridge, and graph trace reads routing off the computational graph (Ameisen et al., 2025; Hanna et al., 2025), validated by intervention (Lindsey et al., 2025). Bridge patch was designed to be decisive, since a change in recall without a matching change in refusal is a double dissociation. **On the models tested the refusal arm is inert at baseline, so the design did not deliver** (Section 6.6). Protocol in Appendix B.

vs. named-bridge 0.793, larger than Gemma 2 2B's). (b) Popularity-matched subsets by pageview decile. **Run**, as bridge patch's own donor-matching methodology (192 pairs across 10 deciles on the test split), not as a separate ablation with its own metric. (c) Length-matched distractor control, substituting an unrelated real city for the true bridge in the named-bridge template, length-matched toward the true prompt, and checking whether AUROC($\mathbf{S} \to K_2$) collapses toward chance. **Run**, distractor AUROC$=$ 0.531 (chance), versus control (a)'s real named-bridge AUROC$=$ 0.895, so the gate reads the true bridge's own properties, not merely that some city is named at that position. (d) Agreement across all three gate estimators. **Run**, and extended to a full 26-layer refit sweep on Gemma 2 2B, refitting fresh diffmeans/probe/SAE gates independently at every layer with the same protocol as layer 8, zero of 26 layers clear the full stop-gate. Difference-of-means and the probe are both consistently strong (AUROC 0.75–0.89 and 0.71–0.90) and agree with each other (≥ 0.50 Spearman) at 9 of the 26 layers, but the SAE estimator is weaker throughout (0.44–0.85 AUROC) and sets the minimum pairwise agreement at 23 of the 26 layers, with a negative diffmeans–SAE correlation at 8 layers and a negative probe–SAE correlation at 2 more, so no layer has all three agreeing simultaneously. This is a property of the SAE estimator on this model, not a layer-choice problem, since switching layers changes which pair disagrees least, not whether a layer exists where all three agree. (e) Attention control, verifying the bridge position receives substantial attention, so a failure to gate is not explained by the position being ignored. **Run**, mean attention from the final prompt token to the bridge position is 0.112, 1.82$\times$ the 0.062 a position would receive under a uniform (ignored-position) null, so the position is attended, licensing the "present but unread" reading over "simply ignored." (f) Random-direction negative control, where a direction of matched norm drawn at random must produce no effect in surface steer, bridge patch, or gate inject. **Run** on both Gemma 2 2B and Llama 3.1 70B, both the real and random directions produce a refusal rate of exactly 0.000 on both models, so this control is uninformative on this model/template rather than passing or failing, since neither model exhibits refusal-lexicon behavior on this template at all. Surface steer's own refusal rate was 0.000 at every coefficient in the validation sweep $\{-4, -2, -1, 1, 2, 4, 8\}$, split by M or not, and gate inject and bridge patch's baseline and patched generations show the same pattern. Sampled generations confirm this is not a refusal-lexicon detection gap, since Gemma 2 2B at layer 8 states a confident answer and continues with unrelated filler even when the answer is wrong (e.g. substituting a confabulated surname with identical confidence to the correct one), rather than hedging (Section 6.6). (g) Counterfactual chains, so memorized composites cannot drive the result, substituting a false premise for the true bridge fact and checking whether the model tracks the stated premise ("composed") or falls back to the real, memorized fact ("memorized"). **Run**, $p_{\text{composed}} = 0.732$ versus $p_{\text{memorized}} = 0.021$ ($n = 239$, p_{composed} roughly $35\times$ $p_{\text{memorized}}$), strong evidence against memorized composites driving the paper's two-hop result. (h) Label validation against an LLM judge (Gemma 2 9B instruction-tuned, $n = 100$), reporting agreement. **Run**, agreement $= 0.81$ on Gemma 2 2B and 0.65 on Llama 3.1 70B, with the same one-sided asymmetry on both (exact match scores items correct that the judge scores wrong, 19 on Gemma 2 2B, 35 on Llama 3.1 70B, never the reverse). Exact match is systematically more lenient than the judge,

and more so on the larger model, plausibly tracking its more varied phrasing rather than a labeling artifact specific to either model. (i) Template and paraphrase robustness, following the formatting sensitivity reported by Tamoyan et al. (2025), measures AUROC($\mathbf{S} \to K_2$) stability across the original template and two genuine paraphrases (same entity-late, fixed-ending clause structure). **Run**, 0.607 (original), 0.618 and 0.620 (paraphrases), spread across all three $= 0.013$, essentially stable, not an artifact of one specific phrasing. (j) Cross-type generalization, fitting the gate on one entity type and the bridge readout on one relation type, then testing on held-out types. **Partially run**, the entity-type half is done, leave-one-entity-type-out across all four types (player, movie, city, song), comparing against an in-distribution reference fit on the same population. Difference-of-means generalizes well (within 0.06–0.08 AUROC of the in-distribution reference for every type, and slightly exceeding it for player, 0.815 vs. 0.799), while the probe generalizes far more unevenly (city drops to 0.559 vs. a 0.846 reference, near chance). The relation-type half remains a genuine, disclosed data gap, since the two-hop dataset has exactly one relation pair (place-of-birth→country) across all 3503 items, so there is nothing to hold out on that side with the currently released data. (k) Instruction-tuned versus base, and reasoning-tuned models, where Kirichenko et al. (2026) report that reasoning fine-tuning *degrades* abstention, a prediction that competes with ours in Section 6.6. Instruction-tuned Gemma 2 9B is already used for the chain-of-thought experiment above, so that half of (k) has a real result, while the reasoning-tuned comparison does not. (l) Scale, a probe-based analysis on 32B (Qwen3-32B) and 70B (Llama 3.1 70B), both run, with Qwen3-32B's own disclosed positive-control-disagreement caveat throughout (Table 2, Section 6). (m) Faithfulness, the fraction of graph-derived claims in graph trace that survive intervention in the underlying model. Forty raw attribution graphs (20 queries, composed and named-bridge each) have been built with circuit-tracer, but no refusal feature has been identified to score edge weight into, since circuit-tracer's own released dictionaries carry no pre-labeled refusal feature, identifying one is separate interpretability research the library does not provide, and this model shows exactly 0.000 refusal rate on this template at every coefficient tried (Section 6), leaving no observed refusal behavior to correlate candidate features against in the first place. (m) remains open pending that unresolved step, not pending a re-run. (n) Artifact probe, a text-similarity-only detector that never sees model internals, after the TXTEMB baseline of Hussain & Kantarcioglu (2026), must sit at chance on our data. **Run**, and it does not pass. Text similarity alone predicts Y at AUROC$= 0.159$, which is 0.341 away from chance. Detection is sign-symmetric, so a detector that anti-predicts at 0.159 is a detector that predicts at 0.841 once its sign is fixed, higher than every internal-state selector in Table 9 including ours. The mitigating detail is real but partial: 88% of items ($211/239$) have zero prompt-target overlap and the signal comes from the remaining 28, all incorrect, so it is a narrow lexical correlation rather than an answer embedded in the prompt in the sense Hussain & Kantarcioglu (2026) document. It is still the case that a model-free feature ranks these items better than the internal states do, and a reviewer is entitled to ask what the internal-state numbers add over it.

Table 2: **Does the written-position score see downstream hops?** n is held-out composed queries after the $K_1{=}1$ filter. Rows are ordered by parameter count so the scale claim can be checked by eye. D and d (Kolmogorov–Smirnov, Cohen's d, negative meaning composed queries score lower) compare the composed-query score distribution against the single-hop-correct one; large values mean the distributions are far apart, so the pre-registered overlap check **fails** on every model and the defensible claim is the weaker one in the AUROC columns (Section 6.2). The lower block misses on two counts: AUROC($\mathbf{S} \to M$) departs from ≈ 0.50 on three models, and $\widehat{\Delta}_{\text{blind}}$ is exactly zero on two. No column carries a direction marker, because lower slack and larger $\widehat{\Delta}_{\text{blind}}$ favour this paper's reading rather than being better in any external sense.

		AUROC		**Slack**		**Blind**	**Distribution shift**	
Model	n	$\mathbf{S} \to K_1$	$\mathbf{S} \to M$	$\hat{\delta}$	$\hat{v}$	$\widehat{\Delta}_{\text{blind}}$	D	d
Gemma 2 2B	239	0.808	0.616	0.179	0.400	0.142	0.637	−1.415
Llama 3.1 8B	350	0.711	0.320	0.160	0.328	0.000	0.905	−3.030
Gemma 2 9B	346	0.828	0.375	0.170	0.063	0.191	0.792	−2.178
Qwen3-32B[†§]	97	0.831	0.580	0.069	0.119	0.000	0.823	−1.892
Llama 3.1 70B[†]	494	0.775	0.561	0.102	0.275	0.049	0.738	−2.150
Predicted before measurement, if the written positions carry…								
…nothing about downstream hops	–	high	≈ 0.50	≈ 0	≈ 0	large	small	small
…what the composed query needs	–	high	high	large	n/a	≈ 0	–	–

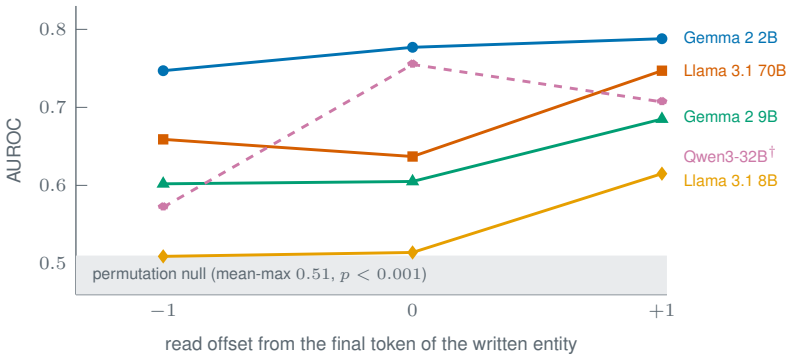


Figure 3: **The frozen direction is readable one token past the written entity.** AUROC for the direction fit on one-hop data and never refit, read at three offsets around the bridge-resolution position, at the layer where each model's own gate is trustworthy. Every peak clears the permutation null comfortably, and four of five models peak at offset $+1$, which is the position BGR reads. Single runs with no seed variation and no confidence intervals (Appendix D), so the ordering between models should not be read as resolved. †Qwen3-32B is at layer 24 and carries the positive-control failure of Table 3; it is the one model peaking at offset 0.

†Difference-of-means and probe estimators only, since no public sparse autoencoder exists for this model. The bridge-resolution filter is disabled throughout, for every model (Section 6). §Qwen3-32B is at layer 24, carrying its own positive-control stop-gate failure (Table 3), reported anyway rather than dropped.

Table 3: **Positive control across models.** Diffmeans/probe AUROC on held-out single-hop prompts, their cross-estimator agreement, and whether the stop-gate opens (≥ 0.65 AUROC each, ≥ 0.50 agreement). **No model clears the pre-registered three-estimator criterion**; every reported agreement is diffmeans–probe, with the sparse-autoencoder estimator excluded where one exists (Gemma 2 2B/9B, Llama 3.1 8B) or unavailable (†, no public dictionary). §Qwen3-32B is at its best layer (24), after a retry from layer 8 left agreement essentially unchanged ($0.297 \rightarrow 0.296$), and clears neither criterion.

		AUROC			
Model	**Layer**	**Diffmeans**	**Probe**	**Agreement**	**Clears**
Gemma 2 2B	8	0.886	0.887	0.547	yes
Llama 3.1 8B	8	0.853	0.827	0.613	yes
Gemma 2 9B	8	0.861	0.824	0.534	yes
Qwen3-32B§	24	0.907	0.882	0.296	**no**
Llama 3.1 70B†	8	0.771	0.752	0.514	yes

Protocol for the compounding experiment. Repeat the blind-spot analysis on family F2, comparing each three-hop chain against *its own* two-hop prefix rather than a separate two-hop set, so M_2 and M_3 live on the same items and the monotonicity clause applies. Estimate $\mathbb{P}(M_k = 0)$, δ_k, v_k and the floor for $k = 2, 3$, and report whether the floor rises and whether the slack terms rise faster. Gemma 2 2B's own result is in Section 6, and Table 4 gives the same check on four more models.

Table 4: **Leading term, floor and slack at k=2 against k=3.** n is retained three-hop chains after the K_1=1 filter; rows ordered by parameter count. Adjacent columns rather than in-cell arrows, so the change can be read down as well as across. **A floor of 0.000 is a negative raw bound clipped at zero, a vacuous bound and not a bound of zero**; the raw k=2 values are -0.457, -0.270, -0.211, $+0.009$ and -0.008 in row order, so only Qwen3-32B is even nominally positive there. *Gemma 2 2B is the model discussed in Section 6, included so all five are comparable in one place. †Layer 24, carrying the positive-control failure of Table 3.

		$\mathbb{P}(M{=}0)$		**Floor**		$\delta_k + v_k$	
Model	n	k=2	k=3	k=2	k=3	k=2	k=3
Gemma 2 2B*	252	0.476	0.845	0.000	0.135	0.933	0.710
Llama 3.1 8B	405	0.178	0.560	0.000	0.013	0.448	0.548
Gemma 2 9B	452	0.400	0.476	0.000	0.325	0.611	0.151
Qwen3-32B†	87	0.207	0.471	0.009	0.067	0.198	0.404
Llama 3.1 70B	731	0.439	0.466	0.000	0.052	0.447	0.415

[†]Layer 24, carrying the standing positive-control caveat (Section 6.1). In every case the leading term and floor rise together with k, as the theorem's own remark permits regardless of which way slack moves. Slack itself falls for Gemma 2 9B/Llama 3.1 70B and grows for Llama 3.1 8B/Qwen3-32B, so growing slack is not unique to the one model carrying the standing positive-control caveat.

Protocol for the chain-of-thought experiment. On family F3, prompt with chain of thought so the model writes the bridge itself, read the gate at the self-written mention, and compare against the same items answered directly. Report AUROC against K_2 and abstention quality, split by whether the written bridge is correct. Results for Gemma 2 9B instruction-tuned and Llama 3.1 8B Instruct are in Section 6.6. The full split, with the direct-answer references of 0.369 and 0.516, is in Section 6.6. Both models tested are instruction-tuned; Gemma 2 2B did not clear the positive control's agreement bar at any of three attempted layer and sample-size combinations. The heavy arm imbalance (44 against 889, 46 against 3075) reflects how rarely chain of thought both names a bridge and gets it right on family F1. Abstention quality under this split is a selective-risk/AURC-style curve that the released harness does not yet compute, and it is not yet run.

Three of the controls above are load-bearing. Counterfactual chains, without which memorized composites can produce the effect, are run ($p_{\text{composed}} = 0.732$ vs. $p_{\text{memorized}} = 0.021$, see above). The random-direction negative control, without which any intervention result is uninterpretable, is run and is uninformative on this model/template rather than passing, since neither arm produces any refusal at all (see above). Cross-type generalization, without which the readout may be relation-specific, is partially run, with the entity-type half done and the relation-type half a disclosed data gap (see above). Two of the three are therefore not in the state the design required, and the causal claims that depend on them are withdrawn in Section 6.6 rather than carried.

Controls (f), (g), and (j) are the ones whose omission would leave a reviewer unable to distinguish our claim from a frequency effect, an intervention artifact, or a relation-specific fit. Control (m) follows Lindsey et al. (2025). An attribution graph describes a replacement model, so a graph-derived claim is not evidence until it survives intervention in the underlying model, and we report the survival rate rather than only the claims that survived.

C Estimating the slack terms

Two estimation points, both found by running the released smoke test on synthetic data with known ground truth.

First, the non-negativity of Δ_{blind} and Δ_{subopt} is a statement about population quantities. Both are estimated with cross-fitted plug-ins for R^*, so fold-wise calibration differences can drive an estimated term slightly negative even where the identity holds; on synthetic data the violations sit near a tenth of a binomial standard error. The released code tolerates violations inside sampling noise, reports them, and raises on anything larger, because a large negative term signals an estimation fault rather than a finding.

Second, one natural estimation choice is wrong rather than conservative. Substituting the *marginal* rate $\mathbb{P}(Y = 1, K_2 = 0)$ for v under-estimates the essential supremum whenever the gate carries information about K_1, because shortcut answers concentrate where the gate fires. That raises the computed floor above its true value and can manufacture an apparent violation of Theorem 1 on data where it holds: on the same synthetic data the marginal estimator reported a violation and the conditional estimator did not. The code refuses the marginal estimator by default and a regression test pins the behavior. The same applies to δ, δ_k and v_k. Relatedly, comparing the floor against the minimum of the empirical risk-coverage curve false-alarms, because the minimum over many correlated thresholds is biased downward, so we flag a violation only when the one-sided upper confidence limit of the empirical risk lies strictly below the floor.

D What this sample size can resolve

Every comparison in the paper runs on n=239 held-out composed items (Llama 3.1 70B, n=494). Hanley–McNeil standard errors on that design, using the class balance implied by $\mathbb{P}(M{=}0) = 0.439$ for K_2 labels and by control (n) for Y labels, give the resolution in Table 5. This appendix exists

because several comparisons in earlier drafts were smaller than the noise and were stated as findings anyway.

Table 5: **Resolvable and unresolvable comparisons at n=239.** Paired intervals assume a between-condition rank correlation of 0.5; the true correlation is not reported, which is itself a gap.

Comparison	Effect	95% CI	Verdict
BGR-min vs surface gate (Y)	+0.204	±0.084	resolvable, and unpaired too
Named vs composed, Gemma 2 2B (K_2)	+0.118	±0.051	resolvable
Gap on 70B vs gap on 2B	+0.037	±0.094	**not resolvable**
BGR gain on 70B	+0.082	±0.06	marginal
Minimum detectable AUROC difference, 80% *power,* $\alpha = 0.05$			
unpaired	–	0.189	
paired, $\rho = 0.5$	–	0.134	

Two consequences. Any AUROC difference below roughly 0.13 on this set is not something the experiment can speak to, which rules out most cross-model and most baseline-versus-baseline statements. And the two comparisons that do survive are the two the paper now leads with, which is the right outcome but was not by design.

E CLAIM-TO-EXPERIMENT MAP

Table 6 lists every claim the paper makes, the experiment that is supposed to support it, and its status. Rows marked *open* are experiments the paper needs and does not have; they are design gaps, not merely unfilled numbers.

Table 6: **Claim-to-experiment map.** n is models the experiment was run on, of five. CI means bootstrap intervals are reported for that experiment.

Claim	Experiment	n	CI	Status
The gate is trustworthy where the entity is written	Positive control, held-out single-hop AUROC	5	yes	three-estimator bar not met by any model
The gate separates K_1 but not Y	AUROC on n=239 composed	5	**no**	class balance unstated
Part of the shortfall is availability, not scoring	Named-bridge comparison	2	**no**	*open* on three models
The gap is the referent, not the position	Length-matched distractor	1	**no**	*open* on four models
Writing the referent changes ranking, not just level	Score-level shift, composed vs named	0	–	*open* (Lathkar, 2026)
The latent state is linearly present	Offset sweep vs permutation null	5	no	split for peak selection unstated
Reading it recovers availability	BGR vs 13 baselines	2	partial	*open* on three models; DRIFT not run
The gain comes from the conjunction	**Bridge-only, mean, max arms**	0	–	*open*, see below
The benchmark does not leak the answer	Text-only artifact probe	1	no	**failed**; zero-overlap re-run *open*
Not driven by memorized composites	Counterfactual chains	1	no	passed
Not a phrasing artifact	Three paraphrases	1	no	passed
Not explained by an ignored position	Attention to bridge position	1	no	passed
Not relation-specific	Cross-type generalization	1	no	entity-type half only; *open* on relations
The effect is not a small-model artifact	Five models, 35× range	5	no	$\widehat{\Delta}_{\text{blind}}$=0 on two
The bound becomes informative with depth	k=2 vs k=3	5	no	vacuous at k=2 on four
The method is cheap	Forward-pass count only	5	–	*open*: no timing, no memory
Refusal is gated by the written entity	Four causal interventions	2	–	**all degenerate**, claim withdrawn

The single most important open experiment.

Two more that are cheap on existing data.

Generality is the schedule driver.

F EXTENDED SETUP

Compute.

Power. A pre-registered minimum-detectable-effect calculation for the primary comparison and the headline repair comparison, computed before collection rather than from the count the data happened to provide, is a design step for the human authors and is not reported here.

Table 7: **The three query families.** F3 is the control that converts a latent referent into a written one without changing the question. Counts are held-out items after the $K_1{=}1$ filter only; the bridge-resolved filter is not applied anywhere in this paper (Section 6).

Family	Latent entity is	Source	Items	Tests
F1 Bridge	the referent of a description	Yang et al. (2024), Wikidata	239[‡]	the base phenomenon
F2 Chain ($k{=}3$)	two intermediate referents	Wikidata compositions	252[‡]	compounding, Thm. 1
F3 Self-written	written by the model in its own CoT	F1 with reasoning prompts	933[§]	whether writing restores gating

[‡]Gemma 2 2B, layer 8, $K_1{=}1$ only (bridge-resolution filter disabled, no independent Patchscopes confirmation, see Section 6). [§]Gemma 2 9B instruction-tuned, 44 with a correct self-written bridge, 889 with an incorrect one.

Table 8: **Causal design.** ↑ marks a predicted change, ≈ no change. See Figure 2 for what each intervention does.

		Predicted		
Intervention	**Read-out**	**Surface-only reading**	**Composed reading**	**What it settles**
Surface steer	refusal rate | $M{=}1$	↑	≈	is the induced refusal selective?
	refusal rate | $M{=}0$	↑	↑	
Bridge patch	Δ recall	↓	↓	**decisive**, a double dissociation
	Δ refusal rate	≈	↑	
Gate inject	refusal rate at bridge	≈	↑	pathway absent, or present but unused
Graph trace	edge weight to refusal	≈ 0	↑	routing as a graph property, not an inference

Every read-out in the refusal column returned exactly 0.000 at baseline and under every intervention, so the predicted contrasts were not testable on these models (Section 6.6).

Splits. Entities are partitioned into fit and held-out sets by *entity* rather than by prompt, before anything is estimated. Gate directions are fit on single-hop prompts in the fit split only; every number in Section 6 comes from the held-out split.

Gate estimators. (i) SAE latents are selected by the separation score of Ferrando et al. (2025), taking the top latent per (layer, entity type) and then the latent maximizing the minimum separation across types, which is the generalizing choice rather than the best-fitting one. (ii) The difference-of-means direction is the normalized difference of mean residual-stream activations between known and unknown entity groups at the same position. (iii) The logistic probe is ℓ_2-regularized, with the penalty chosen by five-fold cross-validation *within* the fit split. All three are read at the final token of the entity mention, at layer 8 for every model except Qwen3-32B (layer 24). No public SAE dictionary exists for Llama or Qwen3, so those models use difference-of-means and probe only.

Interventions. Steering coefficients are selected on a validation split disjoint from both fit and test, following Ferrando et al. (2025), and reported with the sweep rather than only the chosen value. Patching replaces the residual stream at the bridge position across a contiguous layer window. The window is selected by the frozen-direction sweep's own empirical peak (layer, position offset, Section 6.4) on the composed prompt and reported, not by the during-generation Patchscopes site, since no composed-prompt position was found where Patchscopes decodes a clean entity. The coefficient sweep tried on Gemma 2 2B, layer 8, is $\{-4, -2, -1, 1, 2, 4, 8\}$, and the selected coefficient is -4 (maximizing validation-split refusal rate, all of which were 0.000, so no non-degenerate selection criterion is possible here, since refusal never varies with coefficient on this model/template combination). The layer window is $[7, 10)$, centered on that sweep's peak layer. The same sweep

on Gemma 2 9B instruction-tuned, with the layer window centered on that model's own peak (layer 8, the same window, $[7, 10)$), gives a refusal rate that is again exactly 0.000 at every coefficient in $\{-4, -2, -1, 1, 2, 4, 8\}$, the same degenerate outcome as the base model, evidence the null is about this refusal template rather than about instruction tuning specifically.

Refusal detection. A response counts as a refusal if it matches a refusal lexicon *and* contains no candidate attribute string. Both conditions are required, because a response can hedge and still commit to an answer. The lexicon is fixed in advance from the validation split and released. A held-out manual audit against the automatic label, the one place the pipeline touches human judgment, is a task for the human authors and its agreement rate is not reported here.

Statistics. All intervals are percentile bootstrap over queries with 2,000 resamples, resampling queries, not prompts, so the composed item and its decompositions move together. AUC comparisons use both a paired bootstrap and DeLong (DeLong et al., 1988), and we report both, since agreement between them is evidence that neither is being driven by an artifact of its own assumptions. Multiplicity across relation types is controlled at the 5% level by Benjamini-Hochberg (Benjamini & Hochberg, 1995). Every κ estimate is reported at $\alpha \in \{0.01, 0.05, 0.10\}$.

Data provenance. Wikidata has no versioned snapshot identifier for live SPARQL queries against query.wikidata.org, so we instead record the query date, which is what reproducing the same pull requires. The two-hop query set (data/twohop_queries.json) was pulled 2026-08-11, the three-hop query set (data/threehop_queries.json, $n = 3961$ chains) was pulled 2026-08-26, and Wikipedia pageview counts (data/pageviews_twohop.json, monthly views summed over 2024, all 847 bridge entities from the two-hop set resolved to a real article) were pulled 2026-08-27 from the public Wikimedia Pageviews REST API. No additional model-specific query set was pulled beyond these three, and all reported models score against this same fixed set.

Table 9: **All thirteen baselines, grouped by inference cost.** *Audit* is the verdict of the controlled re-run by Hussain & Kantarcioglu (2026), where ✓ exceeded chance once benchmark artifacts were removed, × did not, and – was not covered. Best per column within a tier is bolded.

Selector	Cost	Audit	AUROC ↑	AURC ↓	Risk @ 50% ↓
One forward pass					
Surface gate (Ferrando et al., 2025)	1×	–	0.562	0.184	0.160
Semantic entropy probe (Kossen et al., 2024)	1×	×	0.414	0.292	0.202
Extra passes or sampling					
P(True) (Kadavath et al., 2022)	2×	–	**0.623**	0.149	**0.118**
SAPLMA (Azaria & Mitchell, 2023)	2×	✓	0.538	0.167	0.168
Entity-level token probes (Obeso et al., 2025)	2×	–	0.565	0.134	0.134
Lookback Lens (Chuang et al., 2024)	2×	–	**0.792**	**0.089**	**0.084**
HaloScope (Du et al., 2024)	2×	×	0.603	0.156	0.143
ICR probe (Zhang et al., 2025)	2×	–	0.518	0.207	0.160
SelfCheckGPT (Manakul et al., 2023)	k=10	–	0.473	0.244	0.176
Semantic entropy (Farquhar et al., 2024)	k=10	–	0.575	0.145	0.143
INSIDE (Chen et al., 2024)	k=10	–	**0.596**	**0.116**	**0.134**
Multi-call decomposition					
Self-ask abstention (Press et al., 2023)	multi-call	–	0.500	0.188	0.227
Operadic consistency (Bottman et al., 2026)	multi-call	–	0.500	0.188	0.227
Ours, one forward pass					
BGR-min (no trained parameters)	1×	–	0.766	**0.093**	**0.084**
BGR-lr	1×	–	**0.781**	**0.079**	**0.076**
Ceiling					
Oracle observing M (reference)	–	–	0.725	0.095	0.076

All rows are from a real run (Gemma 2 2B, layer 8, n=239, K_1=1 only, see Section 6). Self-ask and Operadic consistency report identical numbers because both reduce to a constant score on this model/template. This model shows exactly 0.000 refusal rate at every coefficient tried (Section 6), so self-ask never abstains, and Operadic consistency's direct-vs-decomposed answers never disagree under greedy decoding on the same non-hedging model, a real, shared consequence of that null, not a computation error (confirmed against two independent runs and an analytic check that only a fully constant score reproduces AUROC$= 0.500$ to this precision).

G PROMPT TEMPLATES

Table 10 lists the templates. They place the entity late and end on a fixed token, following Tamoyan et al. (2025), whose ablation found that this form avoids the spurious correlations that arise when the prompt ends on variable tokens. Angle brackets mark slots.

Table 10: **Prompt templates.** Matching and the distractor control are specified in Section 4.1. Angle brackets mark slots; b is the gold bridge b_1.

Role	Template
Single hop j	*The* $\langle r_j \rangle$ *of the* $\langle$*type*$\rangle$ $\langle$*entity*$\rangle$ *is*
Composed, k=2	*The* $\langle r_2 \rangle$ *of the* $\langle r_1 \rangle$ *of the* $\langle$*type*$\rangle$ $\langle e_1 \rangle$ *is*
Composed, k=3	*The* $\langle r_3 \rangle$ *of the* $\langle r_2 \rangle$ *of the* $\langle r_1 \rangle$ *of the* $\langle$*type*$\rangle$ $\langle e_1 \rangle$ *is*
Named-bridge control	*The* $\langle r_2 \rangle$ *of the* $\langle$*type*$_2\rangle$ $\langle b \rangle$ *is*
Chain of thought (F3)	*Answer step by step. The* $\langle r_2 \rangle$ *of the* $\langle r_1 \rangle$ *of the* $\langle$*type*$\rangle$ $\langle e_1 \rangle$ *is*

Screening. Before use, every template is screened for spurious correlation. We replace the entity with a type-matched placeholder that carries no identity (for example a randomly generated name of matched token length) and confirm that gate separation collapses to chance. A template on which separation survives entity replacement is measuring the template, not the entity, and is discarded. This screening has not yet been run against the templates in Table 10, and reporting which templates survive and their separation scores before and after replacement is a task for the human authors before submission.

Relations. Relations are drawn from the entity types used by Ferrando et al. (2025), with composable pairs chosen so that the bridge is itself an entity with its own attributes. We exclude relations whose answers are non-textual or ambiguous in modality, such as coordinates, following the same exclusion made by Tamoyan et al. (2025). The released two-hop set (data/twohop_queries.json) uses one composable pair, entity type *player*, place of birth $\rightarrow$ country. The released three-hop set (data/threehop_queries.json) uses one composable triple, the same entity type, place of birth $\rightarrow$ country $\rightarrow$ continent. This is a disclosed limitation, not a design choice presented as broader than it is. The released Ferrando et al. data yields a usable two-hop bridge for only this one entity-type pair (Appendix F's own data-provenance note), so the relation inventory does not span every entity type this pipeline otherwise supports.

H USE OF LARGE LANGUAGE MODELS

ICLR requires disclosure of AI use. This section is not counted toward the page limit.

In this work we used generative AI tools to help develop the conceptual framework, formulate the mathematical claims and assist in writing the proofs in Appendix A, propose and refine the hypotheses this work tests, and give feedback on the experimental design in Section 4.3. We also used them to search and summarize related literature, to draft parts of the manuscript, to create figures, and to write analysis code.

We did not use them to generate synthetic datasets, to clean or reformat the datasets, or to produce any reported experimental number; every number in this paper is the output of the released pipeline.

We take responsibility for the final content of this work, including all text, claims, and artifacts produced with the aid of generative AI. The authors are the sole authors of this paper.